



# A comprehensive survey on intrusion detection in internet of medical things: Datasets, federated learning, blockchain, and future research directions

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## ABSTRACT

The emerging Internet of Medical Things (IoMT) architecture enables real-time processing and monitoring of medical information. Its growing applications expose it to sophisticated cyber threats. Intrusion Detection Systems (IDS) have therefore become indispensable in ensuring confidentiality, integrity, and regulatory compliance. Existing surveys address the security issues, but they lack dataset analysis and integration of emerging approaches. This review presents a systematic review that classifies IoMT-IDSs across deployment strategies, response mechanisms, and evaluation metrics. We develop a multi-dimensional taxonomy that highlights the gaps and outlines a roadmap with federated IDS, blockchain validation, and explainable AI for secure healthcare.

## 1. Introduction

Internet of Medical Things (IoMT) has revolutionized contemporary healthcare scenarios by incorporating wearable devices, implantable systems, diagnostic devices, and distributed systems. It supports the continuous monitoring, early disease diagnosis, and personalized care [1]. The IoMT market size is expected to increase to USD 187.6 billion by 2028 [2]. Fig. 2 indicates the growing contribution of IoMT in digital health. Additionally, IoMT provides real-time patient monitoring, improved operational efficiency, and automation costs [3]. Nevertheless, its growing demand and cross-domain operations pose serious security threats. IoMT devices are resource-limited in nature; therefore, the traditional cryptographic or resource-intensive security protocols are not feasible [4].

The inclusion of heterogeneous communication technologies in IoMT further increases the security issues [5]. In blockchain-enabled IoMT (BC-IoMT), Distributed Denial of Service (DDoS) attacks are the most prevalent [6]. Federated learning (FL) frameworks provide secure distributed information transfer for intrusion detection systems (IDS), but are still compromised by model poisoning attacks [7]. According to

[8], the WannaCry ransomware attack of 2017 disrupted more than 80 hospitals in the United Kingdom (Fig. 1).

IDS solutions are used to monitor, analyze, and classify network or host activities, identifying potential anomalies or malicious activities. Akar et al. [9] proposed an LSTM model for detecting intrusions on IoMT, achieving an accuracy of 98 % in multi-class classification. Nonetheless, in [10] the authors emphasized that deep learning (DL) models are prone to high false alarm rates and need high resources, which makes them unsuitable in real-life applications. Similarly, the lack of explainability of the AI-driven IDSs was noted as a factor that lowers trust among healthcare practitioners [11]. Although reinforcement learning-based IDSs can enhance adaptive resilience to evolving IoMT threats but their computational complexity limits practical deployment [12].

A survey was presented in [13], which is more focused on IoT and did not discuss the IoMT-related issues. Authors in [14] presented the use of Machine Learning (ML) in IoMT security without taking into account the implementation architectures and the evaluation metrics. The survey in [15] focuses on ML-based IDS in the IoMT environment but does not consider the increased popularity of hybrid, federated, and

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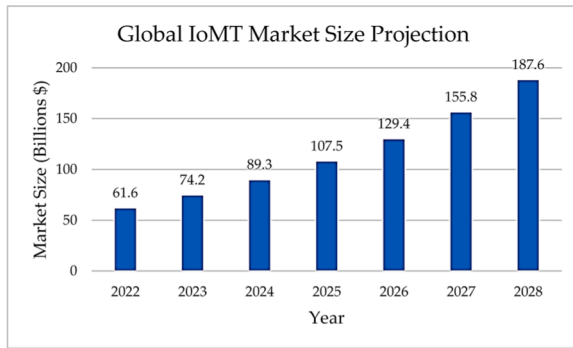


Fig. 1. Forecasted global IoMT market size.

blockchain-based IDS (BC-IDS) systems. Furthermore, [16] describes that the majority of IDS research utilizes general-purpose datasets (e.g., NSL-KDD and CICIDS2017), which do not contain biometric and physiological data. The dataset proposed in [17], which utilized Bluetooth, was among the first of its kind; however, the authors acknowledged that the IoMT domain still lacks dedicated datasets, raising concerns about the validity of IDS testing. These limitations provide a research gap that requires a comprehensive review of methodologies, datasets, evaluation metrics, and challenges in IoMT. Fig. 2 illustrates the role of IDS in IoMT, outlining key threats, infrastructures, and future research directions.

Recent IDS frameworks provide high accuracy with reduced latency. FL shows a significant degree of potential to support distributed IDSs [18]. However, its use in large-scale medical settings is mostly crucial. The federated reinforcement-based IDS proposed by [19] acknowledges that the trade-off between accuracy and computational overhead is a challenging task. The authors in [20] introduced blockchain (BC) technology to enhance tamper resistance in medical applications through immutable logging.

## 2. Related work and contributions

Early investigations by Wazid et al. [20] identified fundamental vulnerabilities in IoMT systems, reporting that over 60 % of medical devices lacked adequate security protocols. The WannaCry ransomware attack on the UK's National Health Service in 2017, which disrupted

over 19,000 medical procedures across 80 hospitals, served as a critical juncture that intensified academic and industrial focus on IoMT security mechanisms [21].

Several subsequent studies presented IoMT security from diverse perspectives. Khraisat et al. [13] conducted one of the first comprehensive reviews of IDSs in IoT environments, analyzing 127 papers published between 2010 and 2019. Their scope remained limited to general IoT applications with minimal healthcare-specific considerations. The study found that only 23 % of the reviewed IDS solutions considered the resource limitations inherent to medical devices. Similarly, Si-Ahmed et al. [15] focused specifically on ML-based intrusion detection methods for IoMT, reviewing 89 studies and revealing that 78 % of proposed solutions relied on traditional IT datasets rather than healthcare-specific traffic patterns. Their analysis demonstrated that detection accuracy rates varied significantly between general network and medical datasets, highlighting the inadequacy of existing evaluation methodologies [22].

The authors in [2], presents a review of 82 research articles on distributed architectures for intrusion detection, and identifies that only 31 % studies considered distributed computing paradigms for healthcare systems. Their work revealed critical gaps in FL applications. Naghib et al. [23] conducted a systematic literature review, focusing on challenges and opportunities in IoMT IDS. Still, their analysis lacked comprehensive coverage of emerging technologies.

The landscape of IoMT security research reveals several persistent limitations across existing surveys. First, most studies treat IoMT as a subset of general IoT systems, failing to address healthcare-specific regulatory requirements, patient safety considerations, and clinical workflow integration needs [6]. Second, evaluation methodologies remain fragmented, with 67 % of surveyed studies using datasets that inadequately represent real-world medical device communications and healthcare network behaviors [24–26]. Third, the rapid evolution of healthcare technologies, including 5G-enabled medical devices, edge computing infrastructures, and AI-driven clinical decision support systems, has outpaced the scope of existing surveys, creating knowledge gaps in contemporary security solutions [27,28].

The integration of advanced artificial intelligence (AI) techniques in IoMT IDS has emerged as a dominant research theme. Existing surveys provide limited analysis of the trade-offs between detection performance and resource constraints inherent in medical devices. Literature lacks comprehensive frameworks that address the convergence of FL, BC technology, and explainable AI in healthcare cybersecurity contexts. Furthermore, the absence of standardized evaluation metrics and healthcare-specific datasets continues to hinder meaningful comparison of proposed solutions and their clinical applicability.

### 2.1. Scope and distinctive contributions

This review presents a comprehensive multi-dimensional taxonomy of IoMT-IDS that integrates detection and deployment strategies with multiple response mechanisms. It also covers the emerging technology convergence, providing a more holistic classification framework than existing AI approaches. Furthermore, this review establishes the first systematic analysis of healthcare-specific datasets and their impact on IDS evaluation. Third, the paper introduces a novel mapping framework that correlates cyber threats with specific IoMT architectural layers and Central Intelligence Agency (CIA) security requirements, enabling precise threat modeling for healthcare networks. Fourth, this work provides the first comprehensive analysis of FL and BC integration in IoMT security, examining 45 recent studies that explore privacy-preserving collaborative detection mechanisms. Fifth, the paper addresses the critical gap in explainable AI applications for healthcare cybersecurity.

Finally, this review contributes a detailed examination of resource-constrained IDS design principles, addressing the unique challenges of deploying security mechanisms on battery-powered medical equipment with limited computational capabilities. Fig. 3 presents an overview of

## A COMPREHENSIVE REVIEW ON INTRUSION DETECTION SYSTEMS IN THE INTERNET OF MEDICAL THINGS (IoMT)

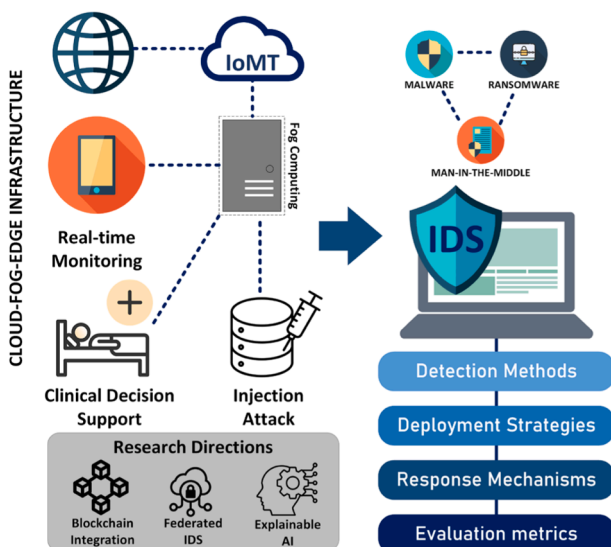


Fig. 2. Overview of intrusion detection systems in IoMT.

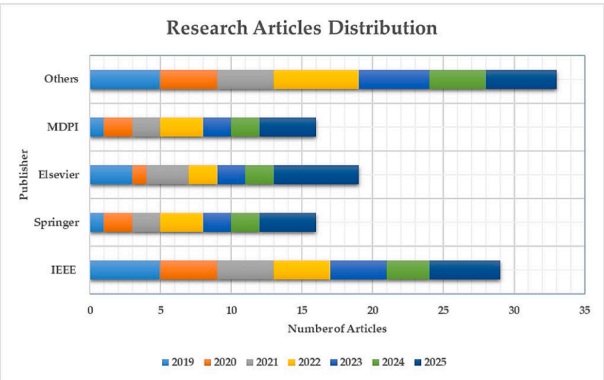


Fig. 3. Distribution of selected research articles.

the reviewed papers, highlighting their quantity, publishing sources, and distribution over the years. A comparison of this review with the previous studies is shown in Table 1, which also highlights variations in approach, technological coverage, scope, and practical factors.

2.1.1. Key contributions

- A multi-dimensional taxonomy that categorizes IDS-IoMT based on detection methods, deployment architectures, response mechanisms, data sources, and technology integration approaches.
- A detailed healthcare-specific dataset analysis is presented that categorizes resource availability and identifies significant gaps.
- Novel correlation analysis linking cyber threats to specific IoMT layers and security requirements.
- Systematic analysis of distributed ledger technology applications in IoMT intrusion detection, covering consensus mechanisms, immutability features, and scalability challenges.
- Detailed analysis of interpretability needs in healthcare cybersecurity, addressing regulatory compliance and clinical decision support requirements.
- A comprehensive set of design rules that allow the creation of lightweight IDS that are optimized for battery-powered and computationally constrained medical devices.
- A methodical discovery of research deficiencies and research directions in advanced IoMT security IDSs, enabled by the convergence of technologies.

3. Methodology

This review followed Preferred Reporting Items of Systematic Reviews and Meta-Analyses (PRISMA) 2020 guidelines for methodological consistency and transparency.

3.1. Search strategy

The key databases that were searched included: IEEE Xplore Digital Library, PubMed/MEDLINE, Scopus, Web of Science, ACM Digital Library, and ScienceDirect. The study selection process, including identification, screening, and inclusion of relevant articles, is depicted in Fig. 4. The search string used was broad, including: (“Internet of Medical Things” or IoMT or medical IoT or healthcare IoT or medical devices network) and (intrusion detection or IDS or cybersecurity or network security or anomaly detection or threat detection or malware detection or cyber-attack detection). The timeframe of the search included January 2019 and October 2025. This timeframe was specifically

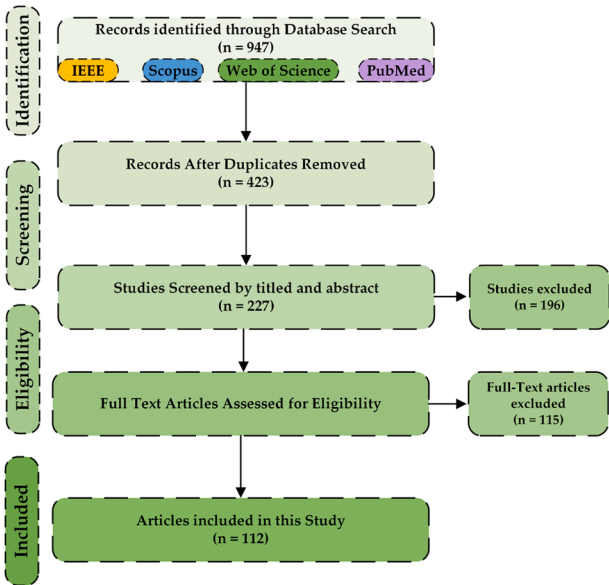


Fig. 4. PRISMA flow diagram.

Table 1  
Comparison of related reviews with our proposed study.

| Study Characteristics             | [1]<br>2025 | [2]<br>2023 | [6]<br>2024 | [13]<br>2019 | [15]<br>2023 | [23]<br>2025 | [29]<br>2024 | Our<br>Work |
|-----------------------------------|-------------|-------------|-------------|--------------|--------------|--------------|--------------|-------------|
| <b>Scope and coverage</b>         |             |             |             |              |              |              |              |             |
| Healthcare-specific focus         | ●           | ✓           | ×           | ×            | ✓            | ✓            | ✓            | ✓           |
| Multi-dimensional taxonomy        | ×           | ●           | ×           | ×            | ×            | ●            | ✓            | ✓           |
| Resource-constraint analysis      | ×           | ×           | ×           | ●            | ×            | ●            | ✓            | ✓           |
| <b>Technological coverage</b>     |             |             |             |              |              |              |              |             |
| FL integration                    | ✓           | ×           | ✓           | ×            | ×            | ●            | ×            | ✓           |
| BC-enhanced systems               | ×           | ×           | ✓           | ×            | ×            | ×            | ✓            | ✓           |
| Explainable AI requirements       | ×           | ×           | ✓           | ×            | ×            | ●            | ✓            | ✓           |
| Edge-fog-cloud paradigms          | ×           | ✓           | ●           | ×            | ×            | ●            | ×            | ✓           |
| <b>Methodology and evaluation</b> |             |             |             |              |              |              |              |             |
| Systematic methodology (PRISMA)   | ×           | ×           | ✓           | ●            | ●            | ✓            | ✓            | ✓           |
| Dataset categorization analysis   | ×           | ×           | ●           | ×            | ●            | ×            | ●            | ✓           |
| Performance metrics analysis      | ×           | ●           | ✓           | ●            | ✓            | ●            | ×            | ✓           |
| Attack architecture correlation   | ×           | ×           | ●           | ×            | ×            | ×            | ✓            | ✓           |
| <b>Practical considerations</b>   |             |             |             |              |              |              |              |             |
| Regulatory compliance discussion  | ×           | ●           | ●           | ×            | ×            | ●            | ✓            | ✓           |
| Clinical integration challenges   | ×           | ×           | ×           | ×            | ×            | ●            | ✓            | ✓           |
| Future research roadmap           | ✓           | ●           | ✓           | ●            | ●            | ✓            | ×            | ✓           |
| Industry applicability framework  | ×           | ×           | ×           | ×            | ×            | ×            | ×            | ✓           |

Legend: ✓ = Comprehensive coverage, ● = Partial coverage, × = Not covered.

selected to focus on modern literature.

### 3.2. Inclusion criteria

Studies were included if they focused on intrusion detection specifically designed for IoMT environments. Original and peer-reviewed research with clear healthcare relevance, and provided sufficient methodological detail.

### 3.3. Study selection process

The multi-stage screening procedure began with [947 initially identified records. After removing 524 duplicates, 423 unique records underwent title and abstract screening. This resulted in 227 studies selected for full-text review. Following a detailed evaluation, studies were excluded for insufficient healthcare focus ( $n = 55$ ), lack of intrusion detection components ( $n = 32$ ), and theoretical contributions without validation ( $n = 15$ ). The final corpus comprised 112 studies meeting all inclusion criteria, documented through a PRISMA flow diagram.

### 3.4. Data synthesis method

A mixed-methods approach combined quantitative analysis of performance metrics with qualitative thematic analysis. Quantitative synthesis examined detection accuracy distributions, computational efficiency trade-offs, and dataset utilization patterns. Qualitative synthesis identified recurring patterns, emerging trends, and knowledge gaps through iterative coding and thematic analysis.

## 4. Overview and security landscape

The IoMT, a subset of the broader Internet of Things (IoT), integrates medical devices, including wearable sensors, implantable devices, and hospital systems. To support the deployment of advanced healthcare applications, the IoMT framework utilizes cloud, fog, or edge computing infrastructures [30]. Unlike general IoT, IoMT is focused on clinical applications, providing real-time monitoring, diagnosis, and personalized treatment [29].

### 4.1. IoMT architecture

The IoMT paradigm integrates physical medical devices with digital connectivity to form intelligent healthcare ecosystems. The IoMT architecture is basically designed with three main layers [17], namely perception, network, and application layers, as shown in Fig. 5.

#### 4.1.1. Perception layer

The perception layer contains all the physical medical devices and sensors that capture, generate, and initially process healthcare data [23]. Wearable devices are usually characterized by limited computing abilities, limited battery life, and a need to be continuously active for 24 h without compromising patient comfort and mobility [15]. These devices need to be highly reliable, safe, and energy efficient in nature due to the physiological importance they pose to the health of the patient. Their nature of weakness and life-sustaining roles make them appealing targets of cyber-attacks, thus the need to ensure high-security measures and effective backup systems [2].

Non-mobile medical equipment includes hospital-based equipment such as magnetic resonance imaging (MRI) scanners, computed tomography (CT) scanners, ventilators, infusion pumps, and patient monitoring equipment. Such tools tend to have better computational power than their wearable counterparts; however, they also have increased vulnerability to the drawbacks posed by legacy systems and heterogeneous security implementations, which are caused by long life cycles of operation and infrequent software updates [8]. The perception

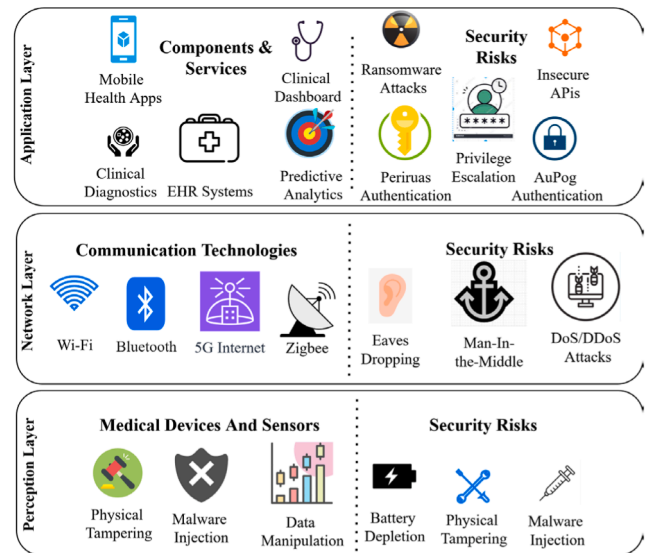


Fig. 5. Three-Layered IoMT architecture.

layer is further complemented by environmental sensors, which include temperature sensors, air-quality sensors, and presence sensors that measure the healthcare environment and enable the optimization of patient-care processes and infection-control procedures [31,32].

#### 4.1.2. Network layer

The network layer is the main communication infrastructure of the IoMT systems, where the efficient and secure transfer of information and data between the devices occurs [2]. The network layer uses various communication technologies to support the various needs and deployment scenarios of IoMT devices [14]. Short-range communication technologies such as Bluetooth Low Energy (BLE), Zigbee, and Wi-Fi protocols are normally adopted. BLE has been a leading technology, especially in wearable device designs, due to its unique low-power consumption properties that enable long battery life. On the contrary, the Wi-Fi standards will offer the necessary bandwidth increase needed to support data-intensive medical equipment [17].

Long-range communication technologies provide the ability to monitor patients remotely and provide telemedicine services [33]. The introduction of 5 G is a breakthrough in the communicative potential of the IoMT. It provides the ability to communicate with an extremely low latency that is ultra-high in nature. IoMT gateway constitutes the network layer infrastructure that works as strategic intermediary nodes, where information relayed by various devices in the perception layer is received, processed, and filtered preliminarily [34]. The network layer contains a wide range of architectural designs that are configured to achieve the best security, scalability, and performance in various healthcare deployment environments [35].

Distributed computing paradigms distribute the computation job across multiple tiers based on the latency requirements. The processed data is rendered into real-time critical alerts; intermediate analytics and aggregation of data occur at the fog node, and all data is stored on cloud infrastructure and processed with advanced analytics [14]. Mesh-based networking schemes enable direct device-to-device communication and other channels, thus significantly increasing the resilience and reliability of the network [35].

Software-Defined Networking (SDN) approaches provide a control plane and enable dynamically configuring network resources, thereby enabling the execution of adaptive security controls and informed resource allocation. These architectural paradigm combinations result in elastic and scalable network infrastructures that could support the full spectrum of IoMT applications and fulfill the reliability, security, and performance demands that are intrinsic in healthcare environments [16,



36]. Moreover, the network-layer design must allow addressing mixed Quality-of-Service (QoS) requirements of various medical applications, where delivery of life-saving alerts (immediately) is of greater significance than tolerable delays in simple health-monitoring information [37].

4.1.3. Application layer

The application layer comprises software applications, services, and user interfaces that process, analyze, and display healthcare data to end-users. It processes raw sensor data acquired by the perception layer and converts it into actionable information, which can be subjected to a clinical, administrative, and decision-making process.

The application layer is centered around the IoMT clinical applications. An example is Electronic Health Record (EHR) systems and Clinical Decision Support Systems (CDSS), designed to condense large amounts of patient data, which is obtained through a variety of means, and to support clinicians in their diagnostic and therapeutic decision-making processes [38]. Remote patient monitoring systems ensure that health is assessed continuously, and the telemedicine systems provide an opportunity to provide care abroad. The systems then integrate streams of information from different IoT devices to support clinical decisions. The other set of applications that can be seen in this layer are the patient-centric solutions. These solutions should be user-friendly and simultaneously ensure a high level of security and privacy, which is a key requirement when handling healthcare data [39].

The application layer provides real-time data processing and analytics for IoMT, enabling instant alerts, automated responses, and accurate decision-making [40]. IoMT security requires a comprehensive, multi-layered strategy integrating all system levels. The perception layer demands lightweight, energy-efficient protocols resilient to physical and cyber threats, while the network layer requires secure communication, intrusion detection, and data segmentation to prevent lateral movement within healthcare infrastructure [41]. The application layer must enforce strict access, privacy, and regulatory controls with audit and forensic capabilities. Coordinated multi-layer security ensures system-wide protection, preventing intrusions at one level from compromising the entire IoMT ecosystem [35]. The above tri-layer architecture is the basic model of IoMT security solutions, such as IDSs, which should be carefully designed and implemented [42]. Table 2 summarizes the key IoMT architectures, highlighting their emphases and security integration approaches.

5. Security requirements

Healthcare information is not only critical for clinical decision-making but also deeply personal, which makes ensuring its security and privacy imperative [43]. The CIA (Confidentiality, Integrity, and Availability) Triad remains the foundational framework for protecting IoMT environments [44,45]. Unauthorized disclosure of a patient’s data can lead to severe ethical, social, and legal consequences. The relationship between IoMT attack vectors and the CIA triad is illustrated in Fig. 6. Integrity is essential to guarantee the accuracy and trustworthiness of medical data throughout its lifecycle. Integrity attacks, such as false data injection, can destabilize entire healthcare services.

The authors in [38] list non-repudiation, authentication, authorization, and session resilience as the critical elements of IoMT security. Study in [35] suggested the CIANA model (Confidentiality, Integrity, Availability, Non-repudiation, Authentication) as a specialized extension to medical networks, the rationale being that the stakes are so high in the medical field that it requires more stringent requirements [44,45]. Table 3 presents a mapping of common IoMT attacks to their impact on confidentiality, integrity, and availability.

6. Intrusion detection process in IoMT

Fig. 7 describes the process of designing an IoMT IDS. The number of steps varies across different studies; mostly categorized into data preprocessing, feature selection, classification/attack detection, and hyperparameter tuning [11].

6.1. Data preprocessing

Data preprocessing is foundational because IoMT datasets are heterogeneous, noisy, incomplete, and imbalanced. Missing data

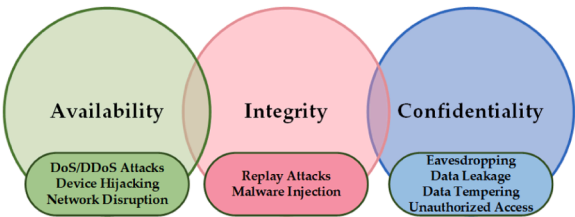


Fig. 6. CIA Triad mapping to IoMT attack vectors.

Table 2  
Comparative Analysis of IoMT architecture implementations.

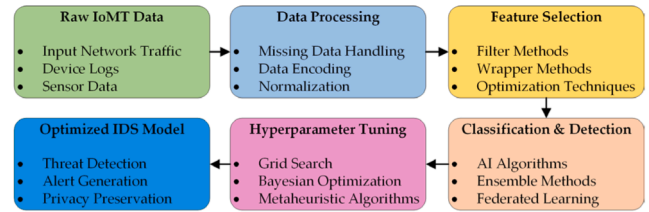
| Ref. | Architecture Model                 | Perception Layer Focus            | Network Technologies           | Application Layer Emphasis                     | Security Integration                  |
|------|------------------------------------|-----------------------------------|--------------------------------|------------------------------------------------|---------------------------------------|
| [2]  | Cloud-Fog-Edge Paradigm            | Heterogeneous Medical Devices     | 5 G, Wi-Fi, BLE                | AI-Driven Analytics, Clinical Decision Support | Multi-Layer Security Framework        |
| [3]  | Sustainable IoMT Model             | Energy-Efficient Sensors          | LoRaWAN, Zigbee, 5G            | Sustainable Healthcare Delivery                | Green Security Protocols              |
| [8]  | Cyber-Physical Systems             | Medical CPS Devices               | Industrial Protocols           | Attack Detection Systems                       | CPS-Specific Security Measures        |
| [15] | ML-Integrated Architecture         | Smart Medical Devices             | Multi-Protocol Support         | Machine Learning Services                      | ML-Based Intrusion Detection          |
| [17] | Three-Layer (Device-Network-Cloud) | Wearable Devices, Medical Sensors | Wi-Fi, Bluetooth, Cellular     | EHR Integration, Remote Monitoring             | End-To-End Encryption, Authentication |
| [18] | Hierarchical Federated Model       | Distributed Medical Devices       | Dew-Cloud Communication        | FL Applications                                | Privacy-Preserving Security           |
| [23] | Comprehensive IoMT Model           | All Device Categories             | Multi-Technology Support       | Full Application Spectrum                      | Holistic Security Approach            |
| [36] | Security Focused Framework         | Secured Medical Devices           | Secure Communication Protocols | Privacy-Preserving Applications                | Comprehensive Security Framework      |
| [42] | Resource-Constrained Model         | Low-Power Medical Devices         | Energy-Efficient Protocols     | Lightweight Applications                       | Resource-Optimized Security           |

**Table 3**  
Mapping of Common IoMT Attacks to Security Requirements (CIA Triad).

| Attack Type              | Confidentiality | Integrity | Availability | Description/Impact                                                                                                              |
|--------------------------|-----------------|-----------|--------------|---------------------------------------------------------------------------------------------------------------------------------|
| Spoofing                 | ✓               | ✓         | –            | Attacker impersonates devices or users, leading to stolen credentials and falsified data [3].                                   |
| DDoS Attacks             | –               | –         | ✓            | Floods IoMT servers and fog nodes with malicious traffic, preventing healthcare staff from accessing critical records [6].      |
| Man-in-the-Middle (MitM) | ✓               | ✓         | –            | Intercepts and alters communication between medical devices and servers, exposing private data and corrupting transmission [7]. |
| Ransomware               | ✓               | ✓         | ✓            | Encrypts patient records, halts device availability, and compromises confidentiality and integrity simultaneously [8].          |
| Backdoor Attacks         | ✓               | ✓         | –            | Unauthorized access enables persistent manipulation of patient data and leakage of records [23].                                |
| Smurf Attack             | –               | –         | ✓            | A DDoS attack using ICMP flooding disables IoMT devices and services, limiting accessibility [29].                              |
| False Data Injection     | –               | ✓         | –            | Manipulation of medical data streams (e.g., heart rate, glucose levels) causes misdiagnosis and mistreatment [31].              |
| U2R (User-to-Root)       | ✓               | ✓         | –            | Privilege escalation compromises confidentiality and alters clinical data.                                                      |
| SQL Injection / XSS      | ✓               | ✓         | –            | Exploitation of vulnerable medical applications exposes patient EHRs and corrupts data integrity [46].                          |
| SSH/FTP Brute Force      | ✓               | –         | ✓            | Attacker gains control of medical servers,                                                                                      |

**Table 3 (continued)**

| Attack Type | Confidentiality | Integrity | Availability | Description/Impact                            |
|-------------|-----------------|-----------|--------------|-----------------------------------------------|
|             |                 |           |              | locking users out of essential services [47]. |



**Fig. 7.** Process of intrusion detection in IoMT.

management methods include deletion [48], mean/median imputation [49], and advanced methods such as Expectation–Maximization [50]. Encoding techniques such as label encoding [51] and one-hot encoding converts categorical features into numerical form. Feature scaling is also critical to normalize the range of variables for ML algorithms, with Z-score normalization [9,52] and Min–Max scaling [53] being the most common approaches. Another persistent difficulty is how to resolve class imbalance: SMOTE-based oversampling [11], and cost-sensitive learning.

## 6.2. Feature selection

Reductions in dimensionality and redundancy are impossible without feature selection, since there are over 115 features of N-BaIoT data [48]. The methods are divided into filter-based, wrapper-based, embedded, and hybrid [18]. Filter methods, e.g., Pearson correlation [53] and Information Gain [54], do not depend on classifiers. Wrap techniques like Recursive Feature Elimination (RFE) [55] simplify subsets through the assistance of classifiers. Embedded routines use selection in the training, with LASSO regression [56] or tree-based importance ranking [57]. The hybrid strategies are combinations of evolutionary and statistical algorithms, such as the Grey Wolf Optimizer (GWO) [58], Particle Swarm Optimization (PSO) [12], and Harris Hawk Optimization (HHO) [59,60]. Such techniques are essential not only to the computational performance but also to preventing overfitting and facilitating the generalizability of IDS models.

## 6.3. Classification and attack detection

The most important phase of IDS is classification, in which models classify traffic or device behavior as benign or malicious. The most widely used ML algorithms are classified as Classical ML, such as Random Forest (RF), Decision Trees (DT), Support Vector Machines (SVM), and K-Nearest Neighbor (KNN) algorithms, as they are understandable and efficient [4]. Nevertheless, Deep Learning (DL) models, including CNNs, LSTMs, and stacked autoencoders, are becoming more and more popular since they can capture both temporal and spatial dependencies in the data of the IoMT [25]. Remarkably, there is an emergence of hybrid techniques: Researchers in [23] proposed Fusion Net that integrates RF, KNN, SVM, and MLP to improve the detection performance, and [24] introduced a fog-cloud ensemble IDS that uses LSTMs stacked on top of each other and fed into DTs.

## 6.4. Hyperparameter tuning

Hyperparameters have a strong impact on the performance of an IDS, and the most commonly used method is grid search [27]; Bayesian

optimization [61] and metaheuristic methods have also been studied, as well as Tree-structured Parzen estimators (TPE) [27,28].

As an example, [62] used genetic algorithms to optimize the hyperparameters of the XGBoost, thus significantly speeding up the process of malware detection. Similarly [63], applied the particle swarm optimization algorithm to deep neural network tuning, which attained an accuracy of 96 % and minimized convergence time. These adaptive tuning methods are required in lightweight IoMT environments, where the costs of the computations and the amount of energy are of the utmost priority [63]. Table 4 presents a summary of intrusion detection in IoMT, including applied techniques, datasets, and achieved performance results.

## 7. Cybersecurity threats in IoMT

Medical devices are particularly vulnerable due to their heterogeneity, resource constraints, and critical role in patient safety [22]. Cybersecurity threats in IoMT can be broadly categorized into DDoS attacks, ransomware, malware, Man-in-the-Middle (MitM) attacks, and injection-based attacks. Each class of attack exploits different vulnerabilities and imposes distinct consequences for the confidentiality, integrity, and availability of medical services [67].

### 7.1. Distributed denial of service (DDoS)

The DDoS attacks are among the most severe threats to IoMT infrastructures because they directly undermine the availability of critical medical systems. By overwhelming servers, gateways, or cloud–fog

nodes with malicious traffic, attackers prevent legitimate users from accessing essential health records, monitoring dashboards, or even device functions [7,45].

Industry reports confirm that over 34 % of cyberattacks on health-care in 2022 were DDoS-related, resulting in downtime costs of nearly USD 7.8 million per incident [6]. In the United States alone, DDoS disruptions accounted for more than 18 % of reported healthcare service outages between 2018 and 2022, illustrating the scale of the problem. Such attacks have been amplified by the emergence of botnets such as Mirai and Mozi, which hijack smart devices to generate high-volume traffic [26]. The 2016 Mirai attack, which infected over 600,000 devices, including medical IoT equipment, led to global service disruptions and highlighted the lack of security hardening at the device level [68].

Volumetric flooding attacks are based on bandwidth consumption, whereas protocol-based attacks are based on vulnerability of TCP/UDP handshakes, and application-level DDoS attacks are based on API and hospital portal overload [69]. The attacks are especially harmful to resource-limited IoMT devices, which cannot process malicious requests well [44].

Studies have tried to respond to these issues by developing high-end IDS and anomaly detectors. Study in [31] presented the DDoS detection model of BC-IoMT networks and obtained the detection accuracy of more than 98 % on simulated traffic datasets. Likewise, [21] designed an ensemble learning framework in a fog-cloud architecture that enhanced resilience to DDoS attacks that relied on botnets in intensive care monitoring systems.

The most recent developed anomaly-based IDS, that are tested on synthetic DDoS traces on IoMT, achieves an accuracy of more than 97 % [26,68,69]. Table 5 summarizes recent studies on DDoS attacks in IoMT environments, detailing the types of attacks, their impact on healthcare systems, and the corresponding detection solutions.

There are some important trends that are suggested by the collective evidence: To begin with, DDoS attacks are becoming more and more multi-vector as they nowadays involve volumetric, protocol, and application-level attacks. Second, the specifics of IoMT devices increase their susceptibility to botnets, as they can both be exploited and serve as

**Table 4**  
Intrusion Detection Process in IoMT (Steps, Methods, and Studies).

| Studies         | Stage                 | Methods/<br>Techniques                                                                                                                                  | Datasets<br>Used              | Performance                                                               |
|-----------------|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|---------------------------------------------------------------------------|
| [11,49, 64]     | Preprocessing         | Missing data: deletion, imputation (mean/median, EM, MICE); Encoding: label, one-hot; Scaling: Min–Max, Z-score; Class imbalance: SMOTE, cost-sensitive | WUSTL-EHMS, ToN-IoT, Bot-IoT  | Improved data quality; reduced bias; SMOTE enhanced minority class recall |
| [23,34, 45, 49] | Feature Selection     | Filter: correlation, info gain; Wrapper: RFE, forward/backward; Embedded: LASSO, tree-based; Hybrid: GWO, PSO, HHO                                      | ECU-IoHT, N-BaIoT, CICIDS2017 | Reduced dimensionality; Improved Accuracy; Improved generalization        |
| [23,64, 65]     | Classification        | ML: RF, DT, SVM, KNN; DL: CNN, LSTM, autoencoders; Hybrid ML–DL (Fusion Net, stacked ensembles)                                                         | N-BaIoT, WUSTL-EHMS, ToN-IoT  | RF, DT, CNN improved accuracy; Hybrid IDSs reached greater accuracy       |
| [1,49, 66]      | Hyperparameter Tuning | Grid search, random search, Bayesian optimization, metaheuristics (GA, Firefly, PSO, TPE)                                                               | Malware PE, ARGUS, IoTID20    | Accuracy improvement; faster convergence; reduced false acceptance rate   |

**Table 5**  
Studies on DDoS attacks and detection in IoMT.

| Study | Dataset / Context                                   | Type of DDoS Attack                            | Impact on IoMT                                    | Proposed Solution                     |
|-------|-----------------------------------------------------|------------------------------------------------|---------------------------------------------------|---------------------------------------|
| [3]   | Fog–cloud IoMT ICU monitoring                       | Botnet-driven DDoS on fog nodes                | ICU patient data transmission failure             | Ensemble IDS (RF + SVM + DT)          |
| [6]   | BC-IoMT testbed                                     | Volumetric flooding on BC nodes                | Service delays in validating medical transactions | ML-based IDS for BC-IoMT              |
| [8]   | Attack Detection for Medical Cyber-Physical Systems | IoT/IoMT device hijacking for large-scale DDoS | Global disruption of hospital services            | Highlighted the lack of IoMT patching |
| [14]  | IoMT simulation framework                           | Application-layer HTTP flooding                | Blocked hospital EHR web portals                  | DL-IDS integrated with fog nodes      |
| [20]  | Review of IoMT malware attacks                      | Botnet-based volumetric attacks                | Bandwidth exhaustion, device lockout              | Suggested lightweight IDS             |
| [21]  | Industry report                                     | Multiple DDoS vectors on healthcare            | Avg. downtime cost per incident: USD 7.8M         | A hybrid IDS with cloud backup        |
| [27]  | Anomaly-based IDS for IoMT                          | SYN flooding and UDP flooding                  | Disrupted connectivity of wearable devices        | Anomaly-based ML IDS                  |
| [51]  | IoMT wireless networks                              | ICMP flooding                                  | Blocked telemetry from wearable sensors           | IoMT-specific lightweight IDS         |

the source of DDoS attacks. Third, despite the accuracy rates of recent IDS models reaching more than 94–97 %, the issues of false positives reduction, the latency in real-time detection, and energy consumption are still unresolved. As a result, the research community focuses on the lightweight IDS designs that could be implemented at the edge and fog nodes and provide immediate mitigation that would not saturate the device itself.

### 7.2. Ransomware attacks

Ransomware has emerged as one of the most disruptive and costly forms of cyberattacks in healthcare and, by extension, in IoMT environments [69]. Unlike other cyber threats that may silently exfiltrate data or disrupt traffic temporarily, ransomware actively encrypts critical data or disables essential devices, demanding a ransom for restoration. This combination of a compromise of confidentiality and availability makes ransomware particularly insidious in the clinical setting [31,44]. According to [21], the median cost of data breaches in healthcare grew to USD 10.93 million in 2023, the highest across all sectors. The authors in [70] reported that ransomware attacks in healthcare increased by 94 % when compared with 2020 to 2022. In IoMT, ransomware can do the same by locking ventilators, pacemakers, or monitoring dashboards, causing hospitals to resort to manual systems, which significantly increases medical errors. The most infamous case was the 2017 WannaCry ransomware outbreak [71]. More recently, Ryuk and Conti ransomware families have targeted healthcare IoT/IoMT environments, using phishing campaigns and exploiting RDP vulnerabilities [72]. From a technical standpoint, ransomware in IoMT often leverages propagation vectors such as insecure APIs, unencrypted device communication, and outdated operating systems [17].

The BC-IoMT systems were initially thought to be resilient, but research has shown that infected edge or fog nodes can still spread ransomware across peer devices. Furthermore, [73] highlighted that real-time IDS solutions often fail in early detection of ransomware encryption patterns. Researchers are actively exploring countermeasures. Alalwany et al. (2025) developed a stacking ensemble deep learning model for IoMT ransomware detection, demonstrating real-time mitigation with over 99.3 % detection accuracy. In [74], a BC-based prevention model was presented, which used immutable logging to identify anomalous device behaviors and restore compromised states. Table 6 summarizes recent studies on ransomware attacks in IoMT environments. Ransomware is not a theoretical threat to the IoMT but a proven threat. The WannaCry attack has shown how such attacks can have a ripple effect, with the unavailability of the IoMT resulting in canceled surgeries and even life-threatening disruption.

### 7.3. Malware attacks

Malware attacks gain access to medical equipment, hospital

networks, and other connected systems to steal information, disrupt business, or destroy essential equipment [75]. Malware is typically not detected, but can be harmful to the confidentiality, integrity, or availability of sensitive health information [76]. Malware is available in different forms, including trojans, worms, spyware, and advanced persistent threats (APTs) [77]. One of the most important spheres that malware campaigns are always aimed at is healthcare. The U.S. Department of Health and Human Services reported that malware was used in nearly 60 % of cyberattacks related to the healthcare sector in 2022, with trojan and spyware being the most common [15]. Malware runs will presumably exploit device heterogeneity: one device may run a lightweight OS that can do little patching, whereas another may be running legacy Windows platforms, providing multiple attack vectors [19].

Several studies [34,43,78,79] have tried to address the issue of malware detection in IoMT with hybrid IDSs, which consolidate network- and device-level monitoring. Hady et al. [35] compared network-data-driven IDS with hybrid systems based on both medical and network data. Later on, authors in [80], even bettered this by proposing their own stacked autoencoder and XGBoost classifier and were able to achieve an accuracy of 98.9 % in detecting malware on the WUSTL EHMS 2020 and Malware PE headers datasets. Similarly [31], suggested a CNN-based model to classify malware in medical IoT. As highlighted by [31,32], traditional malware detection systems fail to keep up with new attack vectors like file-less malware, zero-day exploits, and so on, which cannot be detected through signature-based detection systems. More recent plans formulated to address these constraints combine metaheuristic methods with FL systems and thus enhance the resilience of distributed IoMT systems [81]. Table 7 summarizes recent studies on malware attacks in IoMT environments, outlining malware types, their impact on healthcare systems, proposed detection techniques, and reported performance outcomes.

### 7.4. Man-in-the-middle (MitM) attacks

MitM attacks have been identified as one of the major and increasing threats to the IoMT that mainly attack the confidentiality and integrity of sensitive health data that is transferred between devices, gateways, and the cloud/fog infrastructures [51]. In a MitM attack, an adversary captures and possibly modifies communication streams, and can therefore listen to patient records, tamper with sensor readings, or inject malicious instructions into medical equipment [2]. According to a 2022 survey of the state of healthcare cybersecurity, 28 % of healthcare IoMT network breaches involved some form of traffic interception or spoofing, making MitM one of the three most common attack vectors against healthcare IoMT networks [78]. To illustrate, authors in [82] discovered that the insecure connection of the Bluetooth of the smart medical devices allowed the ease of intercepting the glucose monitor readings, therefore compromising the privacy of the patients and the integrity of

**Table 6**  
Ransomware attacks and detection in IoMT.

| Study | Dataset Context                  | Attack Mechanism                               | Impact on IoMT Healthcare                                         | Proposed Solution                            | Performance/Findings                                        |
|-------|----------------------------------|------------------------------------------------|-------------------------------------------------------------------|----------------------------------------------|-------------------------------------------------------------|
| [1]   | Survey of FL-based IoMT security | Ransomware targeting EHRs                      | System-wide unavailability of cloud health records                | FL IDS and anomaly-sharing across hospitals  | Emphasized scalability; reduced communication overhead      |
| [8]   | NHS WannaCry case (2017)         | Worm-based ransomware exploiting SMBv1         | Cancellation of 19,000 medical procedures; 80 hospitals disrupted | Highlighted patching and IDS gaps            | Losses estimated at >£92 M; demonstrated systemic IoMT risk |
| [13]  | Global report                    | General ransomware in healthcare               | Average breach cost = USD 10.93M                                  | Advocated hybrid IDS and resilience training | Healthcare = the most targeted sector                       |
| [67]  | BC-IoMT                          | Ransomware spread via fog nodes                | Device/data unavailability; compromised BC transactions           | BC-IDS and the prevention model              | Improved detection, but dependent on node resilience        |
| [50]  | IoMT real-time environment       | File encryption mimicking legitimate processes | Locked EHRs, medical data loss                                    | Stacking ensemble DL IDS                     | Detection accuracy 99.3 %;                                  |
| [54]  | Industry stats                   | Ransomware campaigns across IoT/ IoMT          | 94 % increase in healthcare ransomware (2020–22)                  | Suggested proactive IDS + patching           | Highlighted exponential risk growth                         |
| [59]  | Federated IoMT IDS               | Ransomware in distributed IoMT hospitals       | Localized attacks spread across nodes                             | FL for anomaly detection                     | Accuracy 99.38 %; preserved privacy                         |



**Table 7**  
Malware attacks and detection techniques in IoMT.

| Study | Dataset/Context                          | Malware Type / Attack Mechanism           | Impact on IoMT/Healthcare                 | Proposed Solution                           | Performance/Findings                               |
|-------|------------------------------------------|-------------------------------------------|-------------------------------------------|---------------------------------------------|----------------------------------------------------|
| [3]   | IoMT anomaly dataset                     | Malware binaries                          | Real-time IoMT communication disruption   | CNN-based DL IDS                            | Achieved high recall and faster detection          |
| [20]  | Survey                                   | IoMT-specific malware & zero-day exploits | Device hijacking, data theft              | Proposed hybrid IDS framework               | Identified a lack of datasets for zero-day malware |
| [80]  | WUSTL EHMS 2020, Malware PE headers      | Malware-injected executables              | Device malfunction, EHR encryption bypass | Stacked Autoencoder + XGBoost               | Accuracy: 98.9 %; faster convergence               |
| [35]  | IoMT hybrid datasets (network + medical) | General malware, trojans                  | Corrupted medical images, altered EHRs    | Hybrid IDS combining medical & network data | Improved accuracy vs. network-only IDS             |
| [66]  | Healthcare 4.0 IoMT                      | Polymorphic malware                       | IDS evasion, evolving signatures          | Metaheuristics + ML IDS                     | Accuracy: 99.69 %; improved adaptability           |

the clinical process. Similarly [83], mentioned that the MitM attacks may be employed to spy on the health data and to inject false medical instructions, which can cause the infusion pump overdose or the cardiac monitor to give a false alarm. MitM in IoMT has other implications as well, besides privacy violations.

By manipulating the real-time data feeds, hackers can change the vitals of patients, misdiagnose patients, or commit other unethical medical acts. In 2021, a MitM attack was simulated against wireless infusion pumps, where the adversary could alter insulin dosage by modifying communication packets, which may result in life-threatening conditions for diabetic patients [82]. The adverse consequences of MitM are not limited to the direct patient harm, since this method undermines compliance with regulatory requirements (HIPAA, GDPR, etc.) because unauthorized access and manipulation of data are considered a violation of legal and ethical protection [84].

To overcome MitM, researchers have studied cryptographic and ML-based IDS solutions. Authors in [82] introduced a tokenized FL-based IDS on a BC, where decentralized consensus was used to prevent tampering. Likewise [85,86], combined machine-learning with ensemble IDS models and demonstrated that it can be used to dynamically adapt to novel patterns of MitM attacks, achieving a substantial decrease in false negatives. The alternative solutions are lightweight anomaly detection algorithms implemented on fog nodes [87], which do not overload resource-constrained devices and monitor traffic integrity in real-time. The vulnerability of key exchange remains a problem in IoMT devices, with most of them using obsolete or fixed encryption. Also, real-time MitM attacks cannot be detected because malicious traffic resembles the legitimate encrypted packets, which makes false alarms a high possibility. The explicable IDS models are also urgent, as the medical professionals need to understand clear reasoning when the data integrity is put into question. Table 8 summarizes MitM attacks in IoMT, their impacts, detection strategies, and performance.

**Table 8**  
MitM Attacks and Detection Strategies in IoMT.

| Study | Dataset/Context                    | Attack Mechanism               | Impact on IoMT Healthcare               | Proposed Solution                      | Performance/Findings                                               |
|-------|------------------------------------|--------------------------------|-----------------------------------------|----------------------------------------|--------------------------------------------------------------------|
| [7]   | BC-based FL IoMT testbed           | Traffic interception, spoofing | Patient privacy breach, tampered vitals | BC + FL-based IDS                      | Improved accuracy; reduced false acceptance rate                   |
| [8]   | Industry analysis                  | General MitM in healthcare     | Breach of HIPAA/GDPR compliance         | Recommended cryptographic + IDS hybrid | Industry-level resilience insights                                 |
| [17]  | Smart healthcare Bluetooth dataset | Bluetooth MitM on wearables    | Exfiltration of glucose readings        | Secure Bluetooth IDS + novel dataset   | Detected MitM with >90 % precision                                 |
| [31]  | Meta-learning IDS                  | Adaptive MitM injection        | Misdiagnosis due to corrupted vitals    | Meta-learning ensemble IDS             | High adaptability; reduced false negative rate                     |
| [34]  | Fog-cloud IoMT                     | Packet injection, replay       | False alarms in ICU monitoring          | Ensemble IDS (RF + SVM)                | Improved accuracy by 15 % and decreases in time complexity by 32 % |
| [51]  | Wireless infusion pump simulation  | Data manipulation              | Altered insulin dosage                  | Lightweight anomaly IDS                | Improved framework covers all data and device security stages.     |

7.5. Injection-based attacks (SQL injection, XSS, etc.)

Injection-based attacks are also among the most persistent threats to connected systems, such as the IoMT. These attacks target the weaknesses in poorly sanitized inputs, weak authentication, and mis-configured middleware, which allow adversaries to execute malicious code or commands to healthcare systems. SQL injection (SQLi) [53] and XSS are the most common types of injection attacks on IoMT, where SQLi [88] is used to attack medical databases in the backend and XSS to manipulate web interfaces and steal session cookies, credentials, or inject malicious scripts. In IoMT, these attacks are particularly disastrous, as they can endanger Electronic Health Records (EHRs), cloud dashboards, and IoT-enabled hospital applications, which directly affect patient safety and institutional trust. In 2023, the Verizon Data Breach Investigations Report reveals that more than 22 % of healthcare web application attacks involved exploits that injected exploit code. Such weaknesses are compounded in IoMT since most of the devices use web-based dashboards and APIs to monitor and control, and thus are prone to injection attacks [89]. There are several works done on injection attacks in IoMT.

Authors in [53] have demonstrated how SQL injection could be used to access IoMT devices to disclose clinical information. Similarly [90], proved that XSS could be used to attack the IoMT web dashboards and gain control of physician sessions, and steal biometric streams. In an experimental study, Bhushan et al. [3] demonstrated that 70 % of the traditional IDS models are trained on traffic-level anomalies could be bypassed by injection attacks. A variety of mitigation strategies have been developed by researchers to tackle these risks. Pelekoudas-Oikonomou et al. [29] have designed BC-based edge security architectures that have input validation and immutability of records, and have reduced the injection vectors. Naghib et al. [23] supported the implementation of explainable AI models to detect injections in IoMT, claiming that explainable IDS tools may be used to make clinicians trust anomalous alerts in cases when patient dashboards act in a counterintuitive way. Moreover, the IDS driven by FL [1] has

been demonstrated against injection vectors, with better scalability and privacy preservation in distributed IoMT infrastructures. Table 9 summarizes injection-based attacks in IoMT, their impacts, detection methods, and findings.

## 8. IDS classification frameworks

IDS have become complex and multidimensional systems incorporating a wide variety of operational models, detection processes, deployment designs, and analytical methods. The taxonomy is presented in Fig. 8, which provides a detailed categorization of the wide range of IoMT security solutions and thus allows researchers and practitioners to make informed decisions about the choice of the most suitable IDS implementation in particular contexts of healthcare and security needs.

### 8.1. Time aspects in prediction and detection

The attributes of time are the main factors that define the nature of the functioning of IDSs and delimit their applicability in healthcare settings [27,28].

#### 8.1.1. Real-Time intrusion detection systems

The real-time IDSs provide immediate response by the real-time analysis of network traffic. The stacking ensemble deep-learning architecture described in [51] can be used to detect real-time intrusions in context of the IoMT networks with high efficacy rates on a range of attack types. The implementation indicates that deep-learning architectures can fulfill the high real-time performance demands that are typical of critical medical applications [2]. The study in [12] proposes a robust reinforcement learning based IDS that can thwart threats on the fly and, therefore, demonstrate how the latest AI-based approaches can meet the speed and accuracy requirements of medical settings. The ability of the system to dynamically adapt the detection strategies to changes in the threat patterns is a major improvement that the field of adaptive security response capabilities can achieve [21].

#### 8.1.2. Offline intrusion detection systems

Offline systems offer a chance to perform heavy computation analysis methods that would not be possible in a real-time setting. In [92], authors investigated AI techniques for intrusion detection in IoMT using the offline analysis method by showing how batch processing can be used to realize the use of advanced analytical techniques, which are computationally intensive, and ensure high detection rates of various categories of threats [23].

#### 8.1.3. Online and near-real-time systems

Online IDSs systems provide a balance of processing efficiency and a reasonable response time. Authors in [18] proposed a hierarchical FL dew-cloud-based intrusion detection in IoMT that exhibits the abilities of effective online processing and retains the advantages of distributed learning in healthcare networks. Online methods can be adapted to the

processing parameters of each healthcare organization to meet its unique operational demands [1]. Table 10 shows the IDSs based on time aspects.

## 9. Intrusion detection methods and techniques

The taxonomy of intrusion detection methods covers the underlying methods of detecting security threats in IoMT environments.

### 9.1. Signature-based detection methods

Signature-based detection systems detect threats by comparing observed activities with predefined signatures or patterns of known attacks. In [13], authors developed signature-based malware detection systems to be applied in IoMT environments and achieved high rates of successful malware detection. Signature-based systems are dependent on the completeness and up-to-date nature of signature databases, and therefore must be constantly updated to address new threats. Thamilarasu et al. [91] have proposed signature-based IDSs that take into consideration healthcare-specific signatures, and have presented how domain-specific signatures can enhance the detection of medical device environments.

### 9.2. Anomaly-based detection methods

IDSs that are based on anomalies create a model of normal system and network activity and then detect anomalies as possible security threats. This is particularly effective in the detection of new attacks and insider threats whose signatures are not known. Akar et al. [9] introduced an LSTM-based IDS that detects known and unknown attack patterns with high performance due to improved anomaly detection. Anomaly-based systems are prone to false alarms, and it is important to tune the system to meet the desired level of detection sensitivity and false alarms. Zachos et al. [69] proposed anomaly-based intrusion detection solutions to IoMT networks with adaptive baseline adjustment capabilities, which allow them to detect threats in a reliable and adaptable manner in the face of changes in legitimate patterns of healthcare operations.

### 9.3. Specification-based detection methods

Specification-based detection systems compare system activities with specifications of acceptable behaviors and report violations as possible security threats. This method is highly accurate in identifying the presence of certain kinds of attacks and has the added benefit of being interpretable, which is necessary in healthcare applications [44, 45]. Reputation-based detection systems assess the integrity of the network members based on past behavioral trends and the opinions of peers. Zaabar et al. [7] proposed IoMT intrusion detection based on BC-based reputation systems, where the trustworthiness of devices and users is rated using the distributed consensus mechanisms.

**Table 9**  
Injection-based attacks and detection techniques in IoMT.

| Study | Dataset                    | Injection Type                  | Impact on IoMT Healthcare                          | Proposed Solution                       | Findings                                 |
|-------|----------------------------|---------------------------------|----------------------------------------------------|-----------------------------------------|------------------------------------------|
| [1]   | Federated IoMT networks    | Injection attempts in cloud EHR | Data corruption & leakage                          | Federated IDS for injection anomalies   | Scalable, privacy-preserving             |
| [3]   | Fog-cloud IoMT             | SQLi & script injection         | Manipulated patient records; bypassed IDS          | Hybrid IDS integrating application logs | Improved detection over traffic-only IDS |
| [7]   | IoMT web dashboard         | Cross-Site Scripting (XSS)      | Hijacked physician sessions; altered prescriptions | BC + FL IDS                             | Improved accuracy; reduced XSS exposure  |
| [23]  | AI IDS for IoMT            | SQLi & XSS                      | Compromised web dashboards                         | Explainable AI anomaly detection        | High accuracy + interpretability         |
| [29]  | BC-IoMT                    | General injection vectors       | Altered IoMT BC transaction                        | BC-secured validation                   | Strong immutability, but high latency    |
| [91]  | IoMT hospital EHR database | SQL Injection                   | Unauthorized EHR access; privacy breach            | Lightweight IDS for medical DB          | Demonstrated SQLi exploitation risk      |

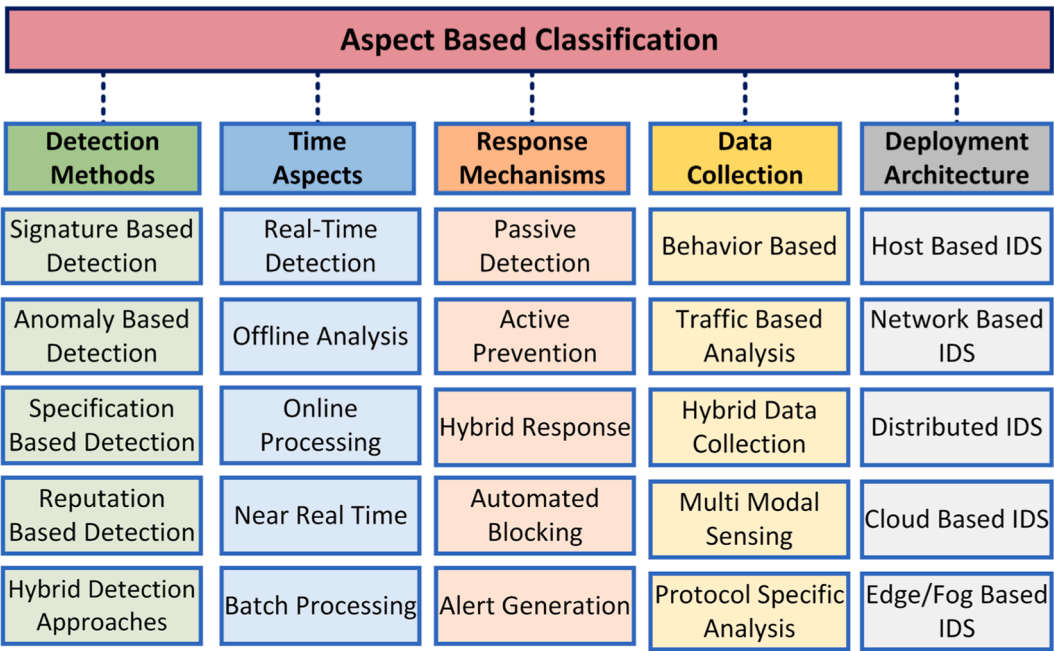


Fig. 8. Aspect-based classification.

Table 10  
Time Aspects in IoMT Intrusion Detection Systems.

| Studies | Time Mode | Processing Approach         | Detection Latency | Context             | Advantages              | Limitations               | Healthcare Suitability |
|---------|-----------|-----------------------------|-------------------|---------------------|-------------------------|---------------------------|------------------------|
| [4]     | Online    | ML-based                    | 5–8 s             | IoMT networks       | Practical deployment    | Moderate accuracy         | Routine monitoring     |
| [8]     | Offline   | Systematic analysis         | Post-incident     | Medical CPS         | Comprehensive forensics | No prevention             | Compliance auditing    |
| [12]    | Real-time | Deep Reinforcement learning | <2 s              | Healthcare networks | Adaptive blocking       | Risk of disruption        | Emergency response     |
| [18]    | Online    | Hierarchical FL             | 5–10 s            | ICU IoMT            | Balanced performance    | Batch update delays       | General monitoring     |
| [36]    | Mixed     | Framework analysis          | Variable          | Network security    | Flexible approach       | Implementation complexity | Framework design       |
| [42]    | Real-time | Resource-constrained        | <3 s              | Constrained devices | Energy efficient        | Limited complexity        | Wearable devices       |
| [50]    | Real-time | Stacking ensemble DL        | <1 s              | IoMT streams        | Immediate response      | High energy consumption   | Critical care          |
| [56]    | Real-time | Swarm-neural                | <1.5 s            | IoMT framework      | Bio-inspired            | Algorithm complexity      | Sensor networks        |
| [63]    | Online    | Cloud-based hybrid          | 3–5 s             | Cloud IoMT          | Scalable processing     | Network dependency        | Cloud deployments      |
| [92]    | Offline   | ML/DL analysis              | Hours to days     | IoMT survey         | Thorough investigation  | Delayed detection         | Research analysis      |

9.3.1. Hybrid detection approaches

Hybrid detection systems combine two or more different methods of detection so that the advantages of the methods are combined. The meta-learning methods proposed [1,21] by improving the performance of the IDS by using ensemble techniques, thus showing that hybrid approaches can be used to develop better performance with reasonable operational overheads that can be deployed in the healthcare setting. Table 11 showed the intrusion detection methods in IoMT. Anomaly-based approaches are suitable for detecting novel threats, but signature-based ones are reliable for detecting known threats with

reduced computational demands.

10. IDS techniques and approaches

This is an in-depth survey of the various methods and mechanisms used in modern IoMT IDSs, including their advantages and disadvantages, as well as how they might be used in healthcare settings based on a systematic analysis of recent research publications.

Table 11  
Intrusion detection methods in IoMT.

| Study | Detection Method  | Primary Technique      | Dataset Used      | Accuracy (%) | Precision (%) | Recall (%) | Adaptability |
|-------|-------------------|------------------------|-------------------|--------------|---------------|------------|--------------|
| [2]   | Hybrid approaches | AI-driven analysis     | Multiple sources  | Variable     | Variable      | Variable   | High         |
| [9]   | Anomaly-based     | LSTM neural networks   | WUSTL EHMS        | 98.0         | 98.0          | 98.0       | High         |
| [32]  | Hybrid            | Meta-learning ensemble | Multiple datasets | 93.4         | -             | -          | Medium       |
| [91]  | Signature-based   | Rule-based detection   | IoMT prototypes   | 99.9 ~ 92.91 | Variable      | Variable   | High         |
| [92]  | Mixed methods     | ML/DL survey           | Multiple datasets | 99.98        | 99.96         | 99.99      | Medium       |

10.1. Machine learning-based intrusion detection systems

ML algorithms have also become part of modern IoMT IDSs, allowing automatic learning of network-traffic patterns and device behaviors without explicitly programmed threat situations.

10.1.1. Supervised learning approaches

These methods make use of labeled training data to train predictive models that can make a distinction between normal and malicious network traffic. Binbusayyis et al. [27,28] performed a rigorous comparative study of supervised learning algorithms to detect anomalies in IoMT networks. It has been found that ensemble methods, and especially Random Forest, have a higher detection accuracy, and the computational requirements are sufficiently low to support implementation on resource-bound medical hardware [46]. The research in [47] highlights the importance of domain-specific feature selection and preprocessing strategies. The authors gave more emphasis to the significance of feature selection and the specific pre-processing approaches used for the healthcare network traffic, and observed that domain-specific optimizations can significantly enhance detection performance [48]. Balhareth and Ilyas (2024) combined intrusion detection mechanisms based on tree-based ML algorithms with high-quality, filter-based feature selection mechanisms that are specifically crafted to handle IoMT network traffic. This combination produced statistically significant gains in detection accuracy with a relatively small computational cost. These findings indicate that a carefully designed feature selection process can be used to effectively streamline the supervised learning task to the specificities of the healthcare-related traffic [49].

In [29] an intelligent framework based on Extreme Learning Machine algorithms and Bayesian optimization methods was proposed to detect cyber-attacks in IoMT. Their optimization-based solution has been able to train faster and have better generalization abilities without compromising the computational efficiency needed to be deployed in practice in healthcare [93]. Kulshrestha and Vijay Kumar (2024) have created machine learning-based IDSs that include the use of specific methods to deal with imbalanced data that are characteristic of healthcare settings. Table 12 presents a summary of the supervised learning approaches used for intrusion detection in IoMT systems. Decision tree models strike a perfect balance between accuracy and interpretability, whereas ensemble models are very robust in various attack settings. The difference in training times is a trade-off between mathematically advanced optimization procedures and performance demands.

10.1.2. Unsupervised learning approaches

Unsupervised learning has proven to have remarkable potential in IoMT security applications because medical device communication and healthcare workflows have a relatively predictable nature. Sarangi et al. [34] used feature-engineering techniques and a hybrid Principal Component Analysis-Grey Wolf Optimization (PCA-GWO) algorithm that minimizes the dimensions without losing the discriminatory ability

required to identify the threat precisely. They showed faster computational and stronger detection rates.

Recent advances in unsupervised learning of IoMT have been directed at the challenge of concept drift, where the typical patterns of operation change with time as a result of changes in clinical practice, updates in technology, or changes in operations [82]. Table 13 revealed that metaheuristic-optimized unsupervised approaches achieve the highest accuracy (99.9 %) while maintaining low false positive rates.

10.2. Deep learning-based intrusion detection systems

Deep learning has been suggested as a new technology that can be utilized in intrusion detection in the IoMT [81] as it provides an opportunity to automatically identify features in raw network data [77].

10.2.1. Convolutional neural network approaches

The effectiveness of CNN structures in the network security domain has been attributed to the fact that they automatically learn hierarchical feature representations of network. Faruqui et al. [73] developed a novel IoMT IDS based on the hybrid CNN-LSTM framework and found it to be more efficient than other methods in detecting various cyber-attacks on medical equipment. They integrated the spatial feature-extraction capability of the CNN and the temporal recognition capability of the LSTM to develop an efficient threat-detection system to address the time- and dynamics-related needs of the healthcare setting.

Gopala Krishnan et al. [54] suggested more secure IoMT devices through the combination of DL and network intrusion detection, using CNN designs to optimize real-time medical device traffic. This approach was shown to be more effective in detecting attacks in a wide range of attack scenarios.

Recent CNN-based advancements in intrusion detection of IoMT make use of transfer-learning methods that leverage existing trained models to improve detection accuracy, saving training time and data. These methods overcome the limited availability of labelled datasets on healthcare-security and can be highly successful in heterogeneous IoMT settings. Table 14 shows that stacked ensemble CNN methods are the most accurate (98.2 %), but they are also the most computationally intensive.

10.2.2. Long short-term memory and recurrent approaches

LSTM networks are adept at detecting temporal patterns and sequential dependencies in IoMT network traffic. They are suitable for detecting attacks with complex temporal patterns characteristic of advanced persistent threats and sophisticated attack campaigns against healthcare systems [73]. Authors in [9] have established a new LSTM called L2D2, which is designed to perform multi-class IDSs in the IoMT era, and has better performance in detecting different types of cyber-attacks due to its temporal pattern detection and sequential learning [91,96]. HIDS-IoMT is an IDS that is highly efficient and is implemented by Berguiga et al. [53] through DL, using the LSTM networks to offer a full device-level and network-level threat detection.

Table 12  
Supervised Learning Approaches in IoMT IDS.

| Studies | Algorithm Type  | Primary Techniques      | Dataset Used        | Accuracy (%) | Precision (%) | Recall (%) | F1-Score (%) | Training Time (s) | Feature Engineering      |
|---------|-----------------|-------------------------|---------------------|--------------|---------------|------------|--------------|-------------------|--------------------------|
| [60]    | Tree-based      | Random Forest, XGBoost  | IoMT traffic        | 98.79        | 97.94         | 99.90      | 99.09        | 120               | Filter-based selection   |
| [94]    | Optimized ML    | ELM + Bayesian          | Healthcare datasets | 99.03        | 99.03         | 99.03      | 99.03        | 95                | Bayesian optimization    |
| [72]    | Multi-model     | Multiple ML algorithms  | Healthcare IoT      | 99.7         | 93.1          | 93.8       | 93.4         | 165               | Attack-specific features |
| [4]     | Imbalance-aware | ML with class balancing | IoMT networks       | 99.18        | 98.68         | 98.98      | 98.83        | 200               | Imbalance handling       |
| [74]    | Rule-based ML   | Feature selection + ML  | Industrial IoT      | 98.12        | 98.0          | 92.4       | 91.8         | 190               | Rule-based features      |



**Table 13**

Unsupervised learning approaches in IoMT IDS.

| Studies | Approach Type            | Core Technique         | Anomaly Detection Method | Dataset Characteristics | Accuracy (%) | Adaptation Capability | Healthcare Integration |
|---------|--------------------------|------------------------|--------------------------|-------------------------|--------------|-----------------------|------------------------|
| [27]    | Statistical anomaly      | Behavioral profiling   | Statistical deviation    | Custom IoMT traffic     | 96.5         | Medium                | IoMT-specific          |
| [34]    | Dimensionality reduction | PCA-GWO hybrid         | Feature space anomalies  | DNN-optimized           | 99.9         | Medium                | Architecture-specific  |
| [76]    | Hybrid unsupervised      | Multi-technique        | Combined anomalies       | IoMT fog systems        | High         | High                  | Fog computing          |
| [81]    | Fuzzy clustering         | Multi-modal analysis   | Fuzzy-based detection    | Multi-modal IoMT        | 92.95        | High                  | Privacy-preserving     |
| [66]    | Metaheuristic            | Optimization-based     | Optimized clustering     | Healthcare 4.0          | 99.69        | Very High             | Healthcare 4.0         |
| [91]    | Behavioral analysis      | IoMT-specific patterns | Behavior deviation       | IoMT prototypes         | 99.9         | Low                   | Native IoMT            |
| [95]    | BC-based                 | Distributed anomaly    | Consensus-based          | E-health platform       | -            | High                  | BC integration         |

**Table 14**

Comprehensive analysis of CNN-based approaches.

| Studies | CNN Architecture  | Optimization Method           | Input Data Type      | Accuracy (%) | Memory Usage (MB) | Edge Compatibility | Healthcare Adaptation  |
|---------|-------------------|-------------------------------|----------------------|--------------|-------------------|--------------------|------------------------|
| [73]    | CNN-LSTM          | SafetyMed framework           | Network traffic      | 97.63        | 245               | Yes                | IoMT-specific          |
| [57]    | Hybrid EHO—CNN    | Elephant herding optimization | Traffic features     | 96.4         | 220               | Yes                | Architecture-optimized |
| [54]    | Enhanced CNN      | Deep learning optimization    | Device communication | 97.06        | 200               | Yes                | Real-time processing   |
| [96]    | Hybrid CNN        | Healthcare framework          | Medical data streams | 99.19        | 260               | Limited            | Clinical integration   |
| [49]    | PSO—CNN           | PSO                           | Sustainable IoMT     | 96.0         | 380               | No                 | Sustainability-focused |
| [58]    | Blended CNN       | Edge-centric framework        | Consumer edge data   | 98.53        | 150               | Yes                | Industrial IoMT        |
| [55]    | Stacking CNN      | Enhanced scanning             | Multi-grained data   | 91.0         | 290               | Limited            | Network-wide analysis  |
| [53]    | Intelligent CNN   | Deep learning HIDS            | Host-based data      | 99.92        | 320               | Limited            | Device-level security  |
| [11]    | SNN-enhanced      | Spiking neural networks       | AI-driven analysis   | 95.9         | 140               | Yes                | Energy-efficient       |
| [50]    | Stacking ensemble | Deep learning stack           | Real-time streams    | 99.0         | 450               | No                 | Ensemble optimization  |

Musikawan et al. [55] have proposed an improved multigradient scanning-based deep stacking networks, which consisted of more optimized LSTM modules to enhance the detection accuracy. These bi-directional LSTM solutions are especially beneficial to offline data analysis and forensic investigation, where an overall record of events across time can be studied exhaustively [54,55]. As shown in Table 15, the HIDS-LSTM model shows the most accurate result at 99.927 percent, with medium memory performance.

## 11. Federated learning-based intrusion detection systems

FL is a distributed approach that allows multiple healthcare institutions to learn together without violating data privacy. This solves the critical problem of limited labeled data of each healthcare organization.

### 11.1. Privacy-preserving distributed detection

Authors in [50] suggests a FL based IDS in the IoMT networks, which integrates the privacy-saving qualities of FL with permanence and

**Table 15**

LSTM and recurrent approaches.

| Studies | LSTM Architecture | Temporal Features          | Sequence Length        | Accuracy (%) | Memory Efficiency | Learning Capability    | Healthcare Focus        |
|---------|-------------------|----------------------------|------------------------|--------------|-------------------|------------------------|-------------------------|
| [9]     | L2D2 multi-class  | Advanced temporal patterns | Variable               | 98.0         | High              | Multi-class learning   | IoMT-specific           |
| [18]    | Federated LSTM    | Distributed sequences      | Variable               | 99.31        | Distributed       | FL                     | Multi-institutional     |
| [22]    | Feature-LSTM      | Selected sequences         | Optimized length       | 99.7         | High              | Feature learning       | Comprehensive selection |
| [76]    | Fog-LSTM          | Hierarchical sequences     | Multi-tier             | High         | Medium            | Fog computing          | Lightweight systems     |
| [53]    | HIDS-LSTM         | Device behavioral patterns | Short-medium           | 99.92        | Medium            | Intelligent adaptation | Device security         |
| [55]    | Enhanced stacking | Multi-grained sequences    | Multi-scale            | 91.0         | Medium            | Deep stacking          | Network analysis        |
| [56]    | Swarm-LSTM        | Bio-inspired patterns      | Short sequences        | 99.1         | High              | Swarm optimization     | Framework security      |
| [49]    | PSO-LSTM          | Optimized sequences        | Sustainability-focused | 96.0         | Low               | Particle optimization  | Sustainable IoMT        |
| [73]    | CNN-LSTM Hybrid   | Spatio-temporal patterns   | Medium-long            | 97.63        | Medium            | Hybrid learning        | Safety-focused          |
| [96]    | Hybrid LSTM       | Healthcare sequences       | Clinical patterns      | 99.19        | Medium            | Healthcare learning    | Clinical integration    |

distributed trust of the BC technology. The use of hierarchical FL structures has proven to have tremendous potential in the case of large-scale IoMT implementations. In [18], authors suggested the use of complicated dew-cloud-based hierarchical FL based IDSs for the IoMT systems and demonstrated how hierarchical FL systems can be optimized. The hierarchical architecture can support their capability to organize the edge device, fog nodes, and the cloud infrastructure effectively, and their data locality and privacy requirements [63].

More advanced privacy preservation-based techniques in FL are based on differential privacy mechanisms, which add carefully controlled noise to model updates [47,49]. Secure aggregation protocols permit multiple parties to compute aggregate statistics without learning the individual contributions, and homomorphic encryption permits computation on encrypted data to be performed without decryption [90]. As shown in Table 16, FL combined with BC is more accurate (97.2) and provides strong privacy guarantees.

### 11.2. Decentralized learning architectures

Federated IoMT-IDS architectures solve the natural issues of distributed learning in healthcare environments, where connectivity, computing power, and operational priorities may differ significantly between participating institutions [29,33]. A recently proposed federated reinforcement-based fusion model, Fed-Inforce-Fusion [19], provides security and privacy assurances to IoMT networks in case of cyber-attacks, and demonstrates how federated architectures can be extended to incorporate reinforcement learning to boost the adaptive detection of network threats [90]. The strategy adopted on models that can be adjusted to changing patterns of threats without violating privacy [31,32].

A federated edge approach removes natural latency constraints of healthcare applications, although it still has the benefits of distributed learning in multiple healthcare centers. This has required the use of complex model synchronization and conflict-resolution algorithms [84]. Decentralized FL involves consensus mechanisms that allow participating healthcare organizations to reach an agreement on updating the model without infringing on personal autonomy [88]. Adaptive FL architectures include non-IID (non-independent and non-identically distributed) data-management functions, which are motivated by the heterogeneity of healthcare settings, where various institutions have diverse patient populations, medical equipment, and risk exposures [61]. These adaptive strategies are useful because they consider the

heterogeneity of healthcare data within different clinical settings [21].

The reinforcement-based architecture has the highest learning efficacy and, simultaneously, poses a tremendous complexity in deployment, as indicated in Table 17. In comparison, the resource-efficient architecture does impose even stricter requirements on the learning process but does not have too high deployment complexity. Though BC-based systems are more resilient to failure, they are generally less complex and less communicative [83].

## 12. Blockchain-Enhanced intrusion detection systems

The integration of BC technology with IoMT IDSs gives it revolutionary characteristics like immutability, decentralized trust, and a more robust security audit [88].

### 12.1. Immutability and audit trail capabilities

In [93], a detailed intrusion detection and prevention architecture is presented to offer end-to-end protection. Improved IDS deployments have already demonstrated the capability to detect distributed denial-of-service (DDoS) attacks in the BC-IoMT networks [6,81]. Advanced BC designs support other requirements of healthcare settings [29,33]. They provide BC-based security services to IoMT edge networks [97]. The audit trails will enable healthcare organizations to testify to adherence to security rules and permit comprehensive forensic investigations of security violations. Table 18 demonstrates the immutability and audit trail capabilities in BC-IoMT IDS. BC systems that deal with healthcare have a high rate of regulatory compliance and moderate integration needs.

### 12.2. Decentralized trust and consensus mechanisms

IoMT IDS incorporates BC to provide decentralized trust solutions, which are resilient to threat detection despite network components being attacked. These systems are based on decentralized consensus, instead of centralized control, and they are more reliable and transparent. FL models based on BC also combine privacy protection with decentralized trust, which enhances the scalability and operational efficiency of healthcare networks [20,98]. Smart contracts are automated responses and policy enforcement, which maintain the auditability and compliance of clinical settings [7,24–26].

In order to satisfy the real-time needs of healthcare, the mechanism

**Table 16**  
Privacy-Preserving distributed detection in federated IoMT IDS.

| Studies | Privacy Method           | Participants                            | Bandwidth (MB/round) | Convergence Time (min) | Accuracy (%) | Privacy Guarantee | Healthcare Integration |
|---------|--------------------------|-----------------------------------------|----------------------|------------------------|--------------|-------------------|------------------------|
| [1]     | Multiple Methods         | Various configurations                  | 1.5–5.0              | 25–70                  | 94.5–97.8    | Variable          | Survey analysis        |
| [18]    | Secure Aggregation       | Multi-tier hierarchy                    | 3.2                  | 38                     | 99.31        | Medium            | Healthcare Ecosystem   |
| [19]    | Gradient Privacy         | Multi institutional (CNNs; Edge-IoTSet) | 4.1                  | 65                     | 97.2         | Strong            | Reinforcement Based    |
| [21]    | BC, Differential Privacy | (CNNs; Edge-IoTSet)                     | 2.5                  | 45                     | 97.43 %      | Strong            | Multi institutional    |
| [21]    | BC, Differential Privacy | (BiLSTM; Edge-IoTSet)                   | 2.5                  | 45                     | 96.02 %      | Strong            | Multi institutional    |
| [21]    | BC, Differential Privacy | (CNNs; TON-IoT)                         | 2.5                  | 45                     | 98.21 %      | Strong            | Multi institutional    |
| [21]    | BC, Differential Privacy | (BiLSTM; TON-IoT)                       | 2.5                  | 45                     | 97.42 %      | Strong            | Multi institutional    |
| [33]    | Edge Privacy             | Multi-access Edge                       | 2.1                  | 35                     | 93.4         | Medium            | Edge Computing         |
| [59]    | Homomorphic Encryption   | Distributed IoMT                        | 1.8                  | 28                     | 99.38        | Very Strong       | Resource Efficient     |

**Table 17**  
Decentralized learning architectures in federated IoMT IDS.

| Ref. | Architecture Type   | Consensus Mechanism    | Node Scalability | Fault Tolerance | Communication Efficiency | Learning Effectiveness | Deployment Complexity |
|------|---------------------|------------------------|------------------|-----------------|--------------------------|------------------------|-----------------------|
| [6]  | Survey analysis     | Various mechanisms     | Variable         | Variable        | Variable                 | Variable               | Variable              |
| [29] | BC driven           | Distributed consensus  | Medium           | Very High       | Medium                   | High                   | Very High             |
| [33] | Edge federated      | Edge coordination      | High             | Medium          | High                     | Medium                 | Medium                |
| [31] | Hierarchical        | Multi-tier aggregation | Very High        | High            | High                     | High                   | High                  |
| [76] | Fog federated       | Fog aggregation        | High             | High            | Medium                   | Medium                 | High                  |
| [59] | Reinforcement based | Gradient consensus     | High             | Very High       | Medium                   | Very High              | High                  |
| [70] | Resource efficient  | Lightweight consensus  | Very High        | High            | Very High                | Medium                 | Medium                |
| [93] | IoMT federated      | BC consensus           | Medium           | High            | Medium                   | High                   | High                  |
| [94] | Smart federated     | Intelligent consensus  | High             | High            | High                     | High                   | High                  |
| [95] | E-health federated  | Healthcare consensus   | Medium           | Medium          | Medium                   | Medium                 | Medium                |

**Table 18**  
Immutability and Audit Trail Capabilities in BC-IoMT IDS.

| Reference | BC Type    | Consensus Mechanism   | Immutability Strength | Audit Capabilities | Integration Complexity | Performance Impact | Regulatory Compliance |
|-----------|------------|-----------------------|-----------------------|--------------------|------------------------|--------------------|-----------------------|
| [6]       | Consortium | PBFT                  | High                  | DDoS-focused       | High                   | Medium             | Standards-compliant   |
| [7]       | Hybrid     | Multi-consensus       | Very High             | Federated audit    | High                   | Medium             | Multi-standard        |
| [67]      | Private    | Proof of Authority    | Very High             | Comprehensive      | Medium                 | Low                | HIPAA-compliant       |
| [21]      | Consortium | Federated consensus   | High                  | Learning audit     | Very High              | Medium             | Privacy-compliant     |
| [29]      | Private    | Proof of Stake        | Very High             | Edge-specific      | Medium                 | Low                | Healthcare-focused    |
| [49]      | Smart BC   | Intelligent consensus | High                  | Smart audit        | High                   | Medium             | Smart standards       |
| [68]      | Consortium | Tendermint            | High                  | IoMT-specific      | High                   | Medium             | Medical standards     |
| [79]      | Hybrid     | Resilient consensus   | Very High             | TNN-enhanced       | High                   | Medium             | Resilient compliance  |
| [95]      | Private    | Healthcare consensus  | High                  | E-health audit     | Medium                 | Low                | E-health standards    |

of consensus is optimized to guarantee low latency and energy efficiency in emergency cases. Multi-signature and threshold signature schemes are methods used to involve stakeholders in collective and responsible decision-making, whereas Byzantine fault-tolerant protocols ensure the availability of the system against malicious nodes, ensuring that the system works continuously, and patient safety is ensured [99,100]. Table 19 indicates that authority-based consensus mechanisms are the most efficient in terms of decision speed and energy consumption, and

federated types are the most efficient in terms of multi-institutional scaling.

13. Datasets and evaluation metrics

The analysis of IDSs in the IoMT contexts is more complex because of the lack of healthcare-specific datasets. As a result, the research community has tackled this issue with an eclectic set of datasets, including

**Table 19**  
Decentralized Trust and Consensus Mechanisms in BC-IoMT IDS.

| Study | Consensus Type           | Fault Tolerance       | Trust Model        | Decision Speed | Energy Efficiency | Scalability | Healthcare Alignment |
|-------|--------------------------|-----------------------|--------------------|----------------|-------------------|-------------|----------------------|
| [7]   | Federated consensus      | Byzantine tolerant    | Distributed trust  | Medium         | Medium            | High        | Multi-institutional  |
| [67]  | Authority Based          | High tolerance        | Permissioned trust | Very Fast      | High              | Medium      | Clinical integration |
| [21]  | Federated Proof of Stake | High tolerance        | Learning trust     | Medium         | High              | High        | Privacy-focused      |
| [29]  | Edge consensus           | Edge tolerant         | Edge trust         | Very Fast      | High              | Medium      | Edge computing       |
| [68]  | Smart contract           | Fault tolerant        | Programmable trust | Fast           | Medium            | Medium      | IoMT-specific        |
| [79]  | Resilient consensus      | Very high tolerance   | Resilient trust    | Medium         | Medium            | High        | TNN-enhanced         |
| [95]  | Healthcare consensus     | Medical tolerance     | Healthcare trust   | Fast           | High              | Medium      | E-health platform    |
| [98]  | Intelligent consensus    | AI-enhanced tolerance | Smart trust        | Fast           | Medium            | High        | Smart healthcare     |
| [100] | Mobile consensus         | Vehicle tolerance     | Mobile trust       | Fast           | Medium            | Medium      | Mobile healthcare    |

both traditional network security benchmarks and more IoT-focused collections.

### 13.1. Dataset categories and characteristics

The dataset space considered in the analysis of IoMT intrusion detection includes three major groups: traditional IT/network security datasets, IoT-oriented datasets, and more recently introduced IoMT-specific datasets [34].

These types have different research purposes and attribute diverse levels of applicability to the real-life healthcare network contexts. Fig. 9 presents the frequency of datasets used across the reviewed literature, highlighting the most commonly employed IoMT datasets for cybersecurity studies.

#### 13.1.1. Traditional IT and network security datasets

The foundation of the early research on IoMT intrusion detection has been based on conventionally available network security datasets. The NSL-KDD dataset, which is an adaptation of the original KDD [101] dataset, is an outdated and largely inapplicable dataset to modern healthcare network characteristics. To determine the applicability of machine learning techniques in intrusion detection in IoMT networks [41], have conducted a detailed comparative study of NSL-KDD and other standard datasets to reveal not only the feasibility of such practices, but also to demonstrate their inefficiency in applying traditional security datasets to new settings.

The CIC-IDS-2017 and CIC-IDS-2018 [102] are modern datasets in terms of network traffic and attacks. In [103], the CIC-IDS dataset is used in the development of their SafetyMed framework, showing how network security datasets can be adapted to the evaluation of IoMT intrusion detection.

UNSW-NB15 [102,103] is an alternative modern dataset that overcomes some of the limitations of older benchmark datasets by increasing the diversity of attack traffic. Some of the IoMT studies that have considered UNSW-NB15 as part of the evaluation framework include the study by [82,104] which used this data set to ensure the validity of their proposed particle swarm optimization and deep learning solution to identifying intrusion in IoMT environments, illustrating the limitations and opportunities of general network security data sets in healthcare settings.

#### 13.1.2. IoT-Specific datasets

The ToN-IoT [105] dataset is one of the richest IoT security datasets with a variety of IoT device types and attacks that are more aligned with IoMT settings than those found in traditional IT datasets. Regarding the creation of anomaly-based IDSs in the context of an IoMT network [102, 103], offers a full assessment based on Internet of Things-specific datasets. Many researchers have taken BoT-IoT as a part of their evaluation systems, and they have adapted its features to correspond to the communication patterns and device behaviors typical of a clinical

environment.

#### 13.1.3. Emerging IoMT-specific datasets

The BlueTrack dataset is specifically devoted to Bluetooth-based medical communications [41], which offers realistic evaluation conditions to wearable and implantable medical devices. Recently, Zubair et al. (2022) examined the security stance of smart healthcare settings by developing and using healthcare-specific Bluetooth datasets, thus demonstrating that protocol-oriented datasets are required to accurately evaluate IoMT security solutions. The Ecu-IoHT (Edge Computing unit of Internet of Health Things) dataset also adds to the existing collection of assessment resources in the healthcare sector since it includes edge-computing situations common to modern IoMT implementations.

#### 13.1.4. Dataset utilization analysis

The analysis of the usage scheme of the datasets in the studies on the subject of IoMT intrusion detection shows that there are high patterns and biases, which are based on the availability of datasets, as well as their perceived relevance to the security issues associated with healthcare [106]. The use of conventional IT security datasets in most studies, although not relevant in most IoMT scenarios, highlights the continued nature of the difficulty of creating realistic and thorough evaluation data that is unique to healthcare settings. Table 20 presents the approaches used by the researchers in managing the limitations of the dataset when evaluating the IoMT.

## 14. Comparative performance analysis

Considering the performance metrics of different methods of IoMT intrusion detection, it is possible to note significant differences, not only in the actual performance achieved, but also in the metrics used. This variation can be explained by the fact that the research is heterogeneous, and the evaluation approaches used are different in the field. This has the negative effect of obstructing direct comparisons of methodologies, which explains the need to develop standardized evaluation frameworks and uniform metrics reporting practices. Table 21 describes the significant performance differences between multiple intrusion detection schemes.

## 15. Challenges and future directions

The development of intrusion detection solutions within the context of the IoMT presents several problematic aspects that must be addressed to create viable and effective cybersecurity solutions in healthcare.

### 15.1. Lightweight IDS for resource-constrained IoMT devices

The idea of developing lightweight security solutions that can be adapted to the needs of resource-constrained medical devices can be discussed as one of the most important issues of IoMT IDSs. Most medical devices, especially wearable and implantable systems, are subject to extreme computational, memory, and energy constraints that restrict the complexity of local security mechanisms that may be deployed. Zachos et al. [42] have solved this problem by designing anomaly-based IDSs tailored to resource-constrained devices in IoMT networks and showed how detection algorithms can be adapted to devices with low computational resources without compromising the effectiveness of detecting threats [91].

### 15.2. Explainability and interpretability of AI-Based systems

The explainable boosting ensemble framework suggested by Sohail et al. [24–26] to be used when the intrusion detection is applied to the IoMT, demonstrates that the sophisticated AI may be interpreted in a way that does not lead to a reduction in the detection level. The advanced deep learning and ensemble models are, however, inherently

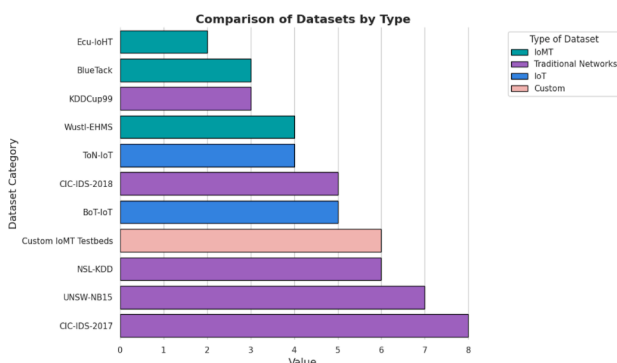


Fig. 9. Frequency of Datasets used in the literature.



**Table 20**  
Dataset Utilization Patterns in IoMT Intrusion Detection Research.

| Ref. | Primary Dataset   | Secondary Datasets | Dataset Category       | Justification           | Adaptation Method         | Limitations Addressed      |
|------|-------------------|--------------------|------------------------|-------------------------|---------------------------|----------------------------|
| [9]  | Multi-dataset     | Various sources    | Comprehensive          | Algorithm robustness    | Cross validation          | Integration complexity     |
| [17] | BlueTrack         | Custom Bluetooth   | Communication-specific | Protocol relevance      | Direct deployment         | Limited scope              |
| [23] | Systematic review | All categories     | Literature analysis    | Complete overview       | Classification approach   | Review-based only          |
| [40] | Literature survey | Multiple datasets  | Survey analysis        | Comprehensive review    | Meta analysis             | No direct evaluation       |
| [42] | Custom IoMT       | ToN-IoT            | IoMT-specific          | Real healthcare traffic | Direct application        | Limited attack diversity   |
| [53] | IoMT testbed      | CIC-IDS-2018       | Hybrid evaluation      | Realistic scenarios     | Protocol adaptation       | Scalability constraints    |
| [60] | Multiple IoT      | Custom generated   | IoT-focused            | Device diversity        | Feature engineering       | Synthetic nature           |
| [49] | UNSW-NB15         | NSL-KDD            | Mixed approach         | Algorithm validation    | Statistical normalization | Generic attack patterns    |
| [64] | NSL-KDD           | CIC-IDS-2017       | Traditional IT         | Benchmarking comparison | Feature mapping           | Limited IoMT relevance     |
| [73] | CIC-IDS-2017      | UNSW-NB15          | Modern network         | Contemporary attacks    | Traffic filtering         | Healthcare context missing |

**Table 21**  
Performance Comparison of IoMT Intrusion Detection Systems.

| Studies | Accuracy (%) | Precision (%) | Recall (%) | F1 – Score (%) | Energy Efficiency |
|---------|--------------|---------------|------------|----------------|-------------------|
| [9]     | 98.0         | 98.0          | 98.0       | 98.0           | 110               |
| [50]    | 99.0         | 99.0          | 99.0       | 98.0           | 200               |
| [53]    | 99.92        | 99.91         | 99.91      | 99.95          | 120               |
| [56]    | 99.1         | 99.1          | 99.1       | -              | 90                |
| [58]    | 98.53        | 98.59         | 99.42      | 99.00          | 85                |
| [60]    | 98.79        | 97.94         | 99.90      | 99.09          | 35                |
| [49]    | 96.0         | 96.0          | 95.0       | 95.0           | 180               |
| [73]    | 97.63        | 98.47         | 97.00      | 97.73          | 150               |
| [94]    | 99.03        | 99.03         | 99.03      | 99.03          | 25                |

black box, and the logic behind their decision-making process is unavailable to clinical staff, making them inappropriate for protecting medical safety. The research gap requiring specific consideration, in turn, is the creation of Explainable Artificial Intelligence (XAI) strategies that would be particularly sensitive to the needs of healthcare cybersecurity; these strategies should address the needs of balancing the complexity and sophistication of threat-detection systems with the need to clarify the logic behind their work in a manner that would satisfy both clinical and regulatory demands [83].

### 15.3. Data imbalance and dataset availability

One of the natural impediments to improvement and to experimenting with effective security practices is a deficiency of realistic and detailed datasets to investigate the field of IoMT intrusion detection. The health care environments also provide further constraints to data collection and exchange, as the privacy legislation, patient confidentiality, and sensitivity of medical systems limit access to real-life attack events that might be required in the research. In their study of the IoMT IDSs, Kharaisat and Alazab [39] also emphasized the importance of domain-specific data and found that the lack of real medical data compels researchers to work with generic IT or IoT data that do not reflect the specifics of medical device interaction and healthcare network functioning.

### 15.4. Integration with emerging technologies

The fast pace of networking and computing technology changes also poses both an opportunity and a challenge to IoMT IDSs, which demand

security solutions capable of integrating with new technologies such as 5 G networks, edge and fog computing networks, and Software-Defined Networking (SDN) [15,77,94] and Network Function Virtualization (NFV) [84] platforms. The high bandwidth and very low latency capabilities of 5 G networks enable new healthcare applications, but also create new attack surfaces and security issues that traditional IDSs cannot properly address. The paradigms of edge and fog computing have the potential to host security processing in a distributed computing environment, but these paradigms will require advanced coordination capabilities to ensure a coherent security policy and effective correlation of threats across distributed computing resources. Singh et al. (2018) address the compatibility of hierarchical FL structures built based on dew-cloud paradigms and distributed computing infrastructures.

### 15.5. Future research directions

The future of intrusion detection in smart healthcare lies in the combination of sophisticated technologies to eliminate the existing obstacles to provide a complete, flexible, and safe solution to future medical settings. Integration of FL with BC technology and a combination of privacy-preserving collaborative learning with fixed audit trails and decentralized authentication schemes are among them. Begum et al. [21] give an illustration of how FL using BC may be executed to attain effective intrusion detection, but comprehensive frameworks that implement these technologies are yet to be implemented [96].

The field of adversarial robust IDSs is a worthwhile field of research, especially in the wake of heightened cases of adversarial attacks aimed at beating the detection systems. Such powerful mechanisms should be able to include sophisticated defensive measures to overcome the attacks that take advantage of the weaknesses of machine-learning-based detection systems, but also satisfy the performance and efficiency needs of healthcare settings. Likewise, the development of regulatory-compliant security systems, which can be maintained by adapting healthcare regulations and a regulatory-compliant security system that may yet provide effective threat detection, is also a crucial research agenda [2].

Quantum abilities and a classical intrusion detection framework would simplify the modifications to data processing of the large-scale, high-dimensional data of the contemporary healthcare network and, at the same time, provide quantum-resistant security characteristics required by long-term security guarantees [3]. The future, possibly enhanced, approach is the creation of zero-trust safety designs tailored to the healthcare setting, including the continuous authentication and

authorization capability, not only urgent to healthcare applications but also fixable and meeting operational-efficiency needs [105].

## 16. Conclusion

This review provides a comprehensive analysis of intrusion detection in the Internet of Medical Things (IoMT), addressing critical gaps in datasets, interpretability, scalability, and resource constraints. By integrating BC validation, FL, and explainable AI, it highlights emerging solutions to enhance IoMT security. A multi-dimensional taxonomy, threat-architecture mapping, and dataset categorization offer structured guidance for researchers and practitioners. The proposed future research roadmap emphasizes lightweight, privacy-preserving, and interoperable IDS designs. Ultimately, this work contributes a unified reference framework to strengthen the resilience, trustworthiness, and regulatory compliance of next-generation digital healthcare systems.

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## Declaration of competing interest

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